

MDM Spread-Anchor Decoding Experiment

Hypothesis

Naive **confidence-based decoding** in masked diffusion models (MDMs) tends toward **left-to-right (L2R)** generation: once a prefix exists, early positions become confident first. The open question is whether **layout** matters---if a few **ground-truth anchor tokens** x_i are revealed far apart in the partial sequence, does a forward pass show higher confidence at **non-local** masked positions (especially between anchors) than revealing the same tokens in a prefix or cluster layout?

This is an **existence / mechanism** claim, not a search for optimal anchors. We test it with **single forward passes** only: fix the prompt and mask state, inject anchors into z , read the confidence landscape on remaining masks. Multi-step decode trajectories wait until existence is settled.

Questions

Primary: do useful far-away anchors exist, and in which domains do they matter most?

Confound: is any spread effect just "more known tokens," or does **where** those tokens sit matter?

If they exist: does a greedy chain pick far placements at later steps and beat a greedy-prefix policy at matched anchor count?

Method

Fixed prompt, mask ratio, and ground-truth completion; one forward pass per partial sequence, batched when possible.

Notation.

- **Ground truth** $x = (x_1, \dots, x_n)$.
- **Partial sequence** z with prompt positions $i \leq r$ fixed to x_i and everything else masked. Let $\mathcal{M}(z)$ denote masked completion indices and $\bar{\mathcal{M}}(z) = \{i : i \leq r\}$ the

prompt.

- **Anchor indices after round k :** $\mathcal{A}_k \subseteq \mathcal{M}(z)$, growing greedily. Still masked: $\bar{\mathcal{A}}_k = \mathcal{M}(z) \setminus \mathcal{A}_k$.
- **Anchored partial $z_{\mathcal{A}_k}$:** $z_i = x_i$ for $i \in \mathcal{A}_k \cup \bar{\mathcal{M}}(z)$ and $z_i = \text{MASK}$ otherwise.
- **Confidence $c_i(z_{\mathcal{A}_k})$** at index $i \in \bar{\mathcal{A}}_k$ (one forward pass on $z_{\mathcal{A}_k}$).
- **Aggregate** $\text{score}(z_{\mathcal{A}_k}) = \sum_{i \in \bar{\mathcal{A}}_k} f(c_i(z_{\mathcal{A}_k}))$.

Write $z = z_\emptyset$ for the baseline (no anchors). Completion length $T = |\mathcal{M}(z)|$.

Metrics.

- **Influence** of anchor candidate p on position q for current state $z_{\mathcal{A}_k}$:
 $I_{p,q}(z, \mathcal{A}_k) = c_q(z_{\mathcal{A}_k \cup \{p\}}) - c_q(z_{\mathcal{A}_k})$ for $p \in \mathcal{M}(z) \setminus \mathcal{A}_k$, and $q \in \mathcal{M}(z) \setminus (\bar{\mathcal{A}}_k \cup \{p\})$.
- **Information Gain:** For $IG(z, \mathcal{A}_k, p, q) = H(z_q | z_{\mathcal{A}_k}) - H(z_q | z_{\mathcal{A}_k \cup \{p\}})$
- **Aggregated Information Gain:** $IG(z, \mathcal{A}_k, p) = \sum_{q \in \bar{\mathcal{A}}_k \setminus (\bar{\mathcal{A}}_k \cup \{p\})} IG(z, \mathcal{A}_k, p, q)$

Greedy anchor selection

Current State: $(k, \mathcal{A}_k)$.

Selection: Select $p \in \bar{\mathcal{A}}_k$ to maximize $IG(z, \mathcal{A}_k, p)$.

Update: $(k + 1, \mathcal{A}_k \cup \{p\})$.

Run k greedy rounds; $k = 1$ is the **existence probe**. Anchor sets grow

$$\mathcal{A}_0 = \emptyset \subset \mathcal{A}_1 \subset \dots \subset \mathcal{A}_k.$$

Cost: $kT - k(k-1)/2 = O(k \cdot T)$ passes per prompt (round 1 alone: $1 + T$; e.g. $k = 3$, $T = 128 \rightarrow 381$ passes vs $\sim 340k$ exhaustive triples).

Additional Layout Control

When greedy picks idiosyncratic sites, hand-specified anchor sets give a clean spread-vs-prefix contrast without search variance. Hold anchor **tokens** x_i fixed (from the greedy chain or round-1 hot spots); permute **indices** only:

Layout	Anchor set $\mathcal{A}_k$
Prefix	first k indices in $\mathcal{M}(z)$
Spread	e.g. maximally separated within $\mathcal{M}(z)$
Cluster-mid	k consecutive center indices in $\mathcal{M}(z)$

Four passes per prompt ($z_{\mathcal{A}_k^{\text{prefix}}}$ vs $z_{\mathcal{A}_k^{\text{spread}}}$). Spread beating prefix on non-local gain at matched k rules out "more tokens" as the sole explanation.

Impact of Data Domain

Run greedy selection (round 1 at minimum; full k where round 1 shows a non-local tail) on ~ 10 -- 20 prompts per bucket:

Domain	Rationale
Template / slot filling	Distant slots may be semantically coupled
Code skeleton	Signature + closing delimiter as natural far anchors
Parallel / list structure	Shared scaffold across repeated clauses
Free-form continuation	Likely weak---L2R prior may dominate

Report per-domain influence-tail strength (round 1) and layout contrast G (Phase 2 or greedy policies). Existence may be domain-conditional.

Non Greedy Scoring (maybe try later)

Information Gain s steps ahead.

Related

- [non-greedy-fim-placeholder-generation](#) --- complementary AR/FIM angle on out-of-order generation; MDM anchors play a similar scaffolding role at decode time.

- [fill-in-the-middle](#) --- prefix--suffix conditioning analogue; spread anchors mimic distant fixed context without a single middle span.